

# MATHEMATICS STANDARD 2024 ALL SETS

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**\*\*CHAPTER: 1 - REAL NUMBERS\*\***

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YEAR: 2024

CHAPTER: 1 - REAL NUMBERS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If  $x = ab^3$  and  $y = a^3b$ , where  $a$  and  $b$  are prime numbers, then  $[HCF(x, y) - LCM(x, y)]$  is equal to:

OPTIONS: (A)  $1 - a^3b^3$  (B)  $ab(1 - ab)$  (C)  $ab - a^4b^4$  (D)  $ab(1 - ab)(1 + ab)$

ANSWER: (D)  $ab(1 - ab)(1 + ab)$

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YEAR: 2024

CHAPTER: 1 - REAL NUMBERS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION:  $(1 + \sqrt{3})^2 - (1 - \sqrt{3})^2$  is:

OPTIONS: (A) a positive rational number. (B) a negative integer. (C) a positive irrational number. (D) a negative irrational number.

ANSWER: (C) a positive irrational number

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YEAR: 2024

CHAPTER: 1 - REAL NUMBERS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The least number which is a perfect square and is divisible by each of 16, 20 and 50, is:

OPTIONS: (A) 1200 (B) 100 (C) 3600 (D) 2400

ANSWER: The correct option is not available in the given options. Full marks may be awarded to every attempt.

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YEAR: 2024

CHAPTER: 1 - REAL NUMBERS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If the HCF of two numbers 65 and 104 is 13. If LCM of 65 and 104 is  $40x$ , then the value of  $x$  is:

OPTIONS: (A) 5 (B) 13 (C) 40 (D) 8

ANSWER: (B) 13

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YEAR: 2024

CHAPTER: 1 - REAL NUMBERS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The sum of the exponents of prime factors in the prime factorisation of 4004 is:

OPTIONS: (A) 5 (B) 4 (C) 3 (D) 2

ANSWER: (A) 5

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YEAR: 2024

CHAPTER: 1 - REAL NUMBERS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The greatest number which divides 281 and 1249, leaving remainder 5 and 7 respectively, is:

OPTIONS: (A) 23 (B) 276 (C) 138 (D) 69

ANSWER: (C) 138

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YEAR: 2024

CHAPTER: 1 - REAL NUMBERS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The LCM of three numbers 28, 44, 132 is:

OPTIONS: (A) 258 (B) 231 (C) 462 (D) 924

ANSWER: (D) 924

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YEAR: 2024

CHAPTER: 1 - REAL NUMBERS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If the product of two co-prime numbers is 553, then their HCF is:

OPTIONS: (A) 1 (B) 553 (C) 7 (D) 79

ANSWER: (A) 1

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YEAR: 2024

CHAPTER: 1 - REAL NUMBERS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If  $3825 = 3^x \times 5^y \times 17^z$ , then the value of  $x + y - 2z$  is:

OPTIONS: (A) 0 (B) 1 (C) 2 (D) 3

ANSWER: (B) 1

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YEAR: 2024

CHAPTER: 1 - REAL NUMBERS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The LCM of the smallest prime number and the smallest odd composite number is:

OPTIONS: (A) 10 (B) 6 (C) 9 (D) 18

ANSWER: (D) 18

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CHAPTER: 1 - REAL NUMBERS

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: Show that the number  $5 \times 11 \times 17 + 3 \times 11$  is a composite number.

OPTIONS: (A) True (B) False (C) Cannot be determined (D) None of the above

ANSWER: Not an MCQ.

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YEAR: 2024

CHAPTER: 1 - REAL NUMBERS

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: Explain why  $7 \times 11 \times 13 + 13$  and  $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$  are composite numbers.

OPTIONS: (A) Both are divisible by 13 (B) Both are divisible by 5 (C) Both have more than two factors (D) Both are even numbers

ANSWER: Not an MCQ.

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YEAR: 2024

CHAPTER: 1 - REAL NUMBERS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: A pair of irrational numbers whose product is a rational number is:

OPTIONS: (A) ( $\sqrt{16}$ ,  $\sqrt{4}$ ) (B) ( $\sqrt{5}$ ,  $\sqrt{2}$ ) (C) ( $\sqrt{3}$ ,  $\sqrt{27}$ ) (D) ( $\sqrt{36}$ ,  $\sqrt{2}$ )

ANSWER: (C) ( $\sqrt{3}$ ,  $\sqrt{27}$ )

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YEAR: 2024

CHAPTER: 1 - REAL NUMBERS

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: Three bells toll at intervals of 9, 12 and 15 minutes respectively. If they start tolling together, after what time will they next toll together?

OPTIONS: (A) 180 minutes (B) 120 minutes (C) 60 minutes (D) 30 minutes

ANSWER: Not an MCQ.

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YEAR: 2024

CHAPTER: 1 - REAL NUMBERS

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: In a school, there are two sections of class X. There are 40 students in the first section and 48 students in the second section. Determine the minimum number of books required for their class library so that they can be distributed equally among students of both sections.

OPTIONS: (A) 240 (B) 120 (C) 480 (D) 80

ANSWER: Not an MCQ.

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YEAR: 2024

CHAPTER: 1 - REAL NUMBERS

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: If  $a = 2^2 \times 3^x$ ,  $b = 2^2 \times 3 \times 5$ ,  $c = 2^2 \times 3 \times 7$  and  $\text{LCM}(a, b, c) = 3780$ , then  $x$  is equal to:

OPTIONS: (A) 1 (B) 2 (C) 3 (D) 0

ANSWER: (C) 3

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YEAR: 2024

CHAPTER: 1 - REAL NUMBERS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The smallest irrational number by which  $\sqrt{20}$  should be multiplied so as to get a rational number, is:

OPTIONS: (A)  $\sqrt{20}$  (B)  $\sqrt{2}$  (C) 5 (D)  $\sqrt{5}$

ANSWER: (D)  $\sqrt{5}$

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**\*\*CHAPTER: 2 - POLYNOMIALS\*\***

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YEAR: 2024

CHAPTER: 2 - POLYNOMIALS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The value of 'a' for which  $ax^2 + x + a = 0$  has equal and positive roots is:

OPTIONS: (A) 2 (B) -2 (C)  $1/2$  (D)  $-1/2$

ANSWER: (D)  $-1/2$

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YEAR: 2024

CHAPTER: 2 - POLYNOMIALS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: Which of the following statements is true for a polynomial  $p(x)$  of degree 3?

OPTIONS: (A)  $p(x)$  has at most two distinct zeroes. (B)  $p(x)$  has at least two distinct zeroes. (C)  $p(x)$  has exactly three distinct zeroes. (D)  $p(x)$  has at most three distinct zeroes.

ANSWER: (D)  $p(x)$  has at most three distinct zeroes.

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YEAR: 2024

CHAPTER: 2 - POLYNOMIALS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The zeroes of a polynomial  $x^2 + px + q$  are twice the zeroes of the polynomial  $4x^2 - 5x - 6$ . The value of  $p$  is:

OPTIONS: (A)  $-5/2$  (B)  $5/2$  (C)  $-5$  (D)  $10$

ANSWER: (C)  $-5$

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YEAR: 2024

CHAPTER: 2 - POLYNOMIALS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The zeroes of the quadratic polynomial  $2x^2 - 3x - 9$  are:

OPTIONS: (A)  $3, -3/2$  (B)  $-3, -3/2$  (C)  $-3, 3/2$  (D)  $3, 3/2$

ANSWER: (A)  $3, -3/2$

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YEAR: 2024

CHAPTER: 2 - POLYNOMIALS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If  $\alpha$  and  $\beta$  are the zeroes of the polynomial  $p(x) = kx^2 - 30x + 45k$  and  $\alpha + \beta = \alpha\beta$ , then the value of  $k$  is:

OPTIONS: (A)  $-2/3$  (B)  $-3/2$  (C)  $3/2$  (D)  $2/3$

ANSWER: (D)  $2/3$

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YEAR: 2024

CHAPTER: 2 - POLYNOMIALS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: What should be added from the polynomial  $x^2 - 5x + 4$ , so that  $3$  is the zero of the resulting polynomial?

OPTIONS: (A)  $1$  (B)  $2$  (C)  $4$  (D)  $5$

ANSWER: (B)  $2$

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YEAR: 2024

CHAPTER: 2 - POLYNOMIALS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: A quadratic polynomial, one of whose zeroes is  $2 + \sqrt{5}$  and the sum of whose zeroes is 4, is:

OPTIONS: (A)  $x^2 + 4x - 1$  (B)  $x^2 - 4x - 1$  (C)  $x^2 - 4x + 1$  (D)  $x^2 + 4x + 1$

ANSWER: (B)  $x^2 - 4x - 1$

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YEAR: 2024

CHAPTER: 2 - POLYNOMIALS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If  $\alpha$  and  $\beta$  are zeroes of the polynomial  $5x^2 + 3x - 7$ , the value of  $1/\alpha + 1/\beta$  is:

OPTIONS: (A)  $3/7$  (B)  $3/5$  (C)  $-3/7$  (D)  $5/7$

ANSWER: (A)  $3/7$

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YEAR: 2024

CHAPTER: 2 - POLYNOMIALS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: What should be subtracted from the polynomial  $x^2 - 16x + 30$ , so that 15 is the zero of the resulting polynomial?

OPTIONS: (A) 30 (B) 14 (C) 15 (D) 16

ANSWER: (C) 15

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YEAR: 2024

CHAPTER: 2 - POLYNOMIALS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If  $\alpha$  and  $\beta$ , ( $\alpha > \beta$ ), are the zeroes of the polynomial  $-x^2 + 8x + 9$ , then  $(\alpha - \beta)$  is equal to:

OPTIONS: (A) -10 (B) 10 (C)  $\pm 10$  (D) 8

ANSWER: (B) 10

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YEAR: 2024

CHAPTER: 2 - POLYNOMIALS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The value of 'a' for which the quadratic equation  $ax^2 + x + a = 0$  has equal and positive roots is:

OPTIONS: (A) 2 (B) -2 (C)  $1/2$  (D)  $-1/2$

ANSWER: (D)  $-1/2$

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YEAR: 2024

CHAPTER: 2 - POLYNOMIALS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If  $\alpha$  and  $\beta$  are zeroes of the polynomial  $2x^2 - 9x + 5$ , then value of  $\alpha^2 + \beta^2$  is:

OPTIONS: (A)  $1/4$  (B)  $61/4$  (C) 1 (D)  $71/4$

ANSWER: (B)  $61/4$

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YEAR: 2024

CHAPTER: 2 - POLYNOMIALS

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: If  $\alpha, \beta$  are zeroes of the polynomial  $p(x) = 21y^2 - y - 2$ , then find the value of  $(1 - p) \cdot (1 - q)$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 2 - POLYNOMIALS

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: Find the zeroes of the quadratic polynomial  $x^2 - 15$  and verify the relationship between the zeroes and the coefficients of the polynomial.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 2 - POLYNOMIALS

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: (a) Find the zeroes of the polynomial  $4x^2 + 4x - 3$  and verify the relationship between zeroes and coefficients of the polynomial.

OR

(b) If  $\alpha$  and  $\beta$  are the zeroes of the polynomial  $x^2 + x - 2$ , then find the value of  $\alpha/\beta + \beta/\alpha$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 2 - POLYNOMIALS

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: (a) Find the zeroes of the polynomial  $4x^2 + 4x - 3$  and verify the relationship between zeroes and coefficients of the polynomial.

OR

(b) If  $\alpha$  and  $\beta$  are the zeroes of the polynomial  $x^2 + x - 2$ , then find the value of  $\alpha/\beta + \beta/\alpha$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 2 - POLYNOMIALS

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: If  $\alpha, \beta$  are zeroes of the polynomial  $p(y) = y^2 - 5y + 3$ , then find the value of  $\alpha^4\beta^3 + \alpha^3\beta^4$ .

OPTIONS: (A) 135 (B) 270 (C) 45 (D) 15

ANSWER: (A) 135

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YEAR: 2024

CHAPTER: 2 - POLYNOMIALS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If  $\alpha$  and  $\beta$  are zeroes of the polynomial  $2x^2 - 9x + 5$ , then value of  $\alpha^2 + \beta^2$  is:

OPTIONS: (A)  $1/4$  (B)  $61/4$  (C) 1 (D)  $71/4$

ANSWER: (B)  $61/4$

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YEAR: 2024

CHAPTER: 2 - POLYNOMIALS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: One of the zeroes of the polynomial  $p(x) = kx^2 - 9x + 3$  is  $(-3/2)$ . The value of  $k$  is:

OPTIONS: (A)  $22/3$  (B)  $-14/3$  (C)  $14/3$  (D)  $-22/3$

ANSWER: (D)  $-22/3$

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YEAR: 2024

CHAPTER: 2 - POLYNOMIALS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The zeroes of a polynomial  $x^2 + px + q$  are twice the zeroes of the polynomial  $4x^2 - 5x - 6$ . The value of  $p$  is:

OPTIONS: (A)  $-5/2$  (B)  $5/2$  (C)  $-5$  (D)  $10$

ANSWER: (C)  $-5$

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YEAR: 2024

CHAPTER: 2 - POLYNOMIALS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The value of ' $k$ ' for which the quadratic equation  $(k+1)x^2 - 6(k+1)x + 3(k+9) = 0$ ,  $k \neq -1$  has real and equal roots is:

OPTIONS: (A)  $3$  (B)  $-3$  (C)  $0$  (D)  $1$

ANSWER: Not a single MC option.

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YEAR: 2024

CHAPTER: 2 - POLYNOMIALS

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: Find the zeroes of the polynomial  $p(x) = x^2 + 4/3x - 4/3$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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\*\*CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES\*\*

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YEAR: 2024

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The value of 'p' for which the equations  $px + 3y = p - 3$ ,  $12x + py = p$  has infinitely many solutions is:

OPTIONS: (A) -6 only (B) 6 only (C)  $\pm 6$  (D) Any real number except  $\pm 6$

ANSWER: (B) 6 only

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YEAR: 2024

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The pair of linear equations  $x + 2y + 5 = 0$  and  $-3x = 6y - 1$  has:

OPTIONS: (A) unique solution (B) exactly two solutions (C) infinitely many solutions (D) no solution

ANSWER: (D) no solution

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YEAR: 2024

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If  $ax + by = a^2 - b^2$  and  $bx + ay = 0$ , then the value of  $x + y$  is:

OPTIONS: (A)  $a^2 - b^2$  (B)  $a + b$  (C)  $a - b$  (D)  $a^2 + b^2$

ANSWER: (C)  $a - b$

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YEAR: 2024

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: Which out of the following type of straight lines will be represented by the system of equations  $3x + 4y = 5$  and  $6x + 8y = 7$ ?

OPTIONS: (A) Parallel (B) Intersecting (C) Coincident (D) Perpendicular to each other

ANSWER: (A) Parallel

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YEAR: 2024

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If a pair of linear equations in two variables is consistent, then the lines represented by the two equations are:

OPTIONS: (A) always intersecting (B) parallel (C) always coincident (D) intersecting or coincident

ANSWER: (D) intersecting or coincident

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YEAR: 2024

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The pair of linear equations  $y = 0$  and  $y = -7$  have:

OPTIONS: (A) exactly one solution (B) two solutions (C) infinitely many solutions (D) no solution

ANSWER: (D) no solution

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YEAR: 2024

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: Two lines are given to be parallel. The equation of one of these lines is  $5x - 3y = 2$ . The equation of the second line can be:

OPTIONS: (A)  $-15x - 9y = 5$  (B)  $15x + 9y = 5$  (C)  $9x - 15y = 6$  (D)  $-15x + 9y = 5$

ANSWER: (D)  $-15x + 9y = 5$

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YEAR: 2024

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: (a) If  $2x + y = 13$  and  $4x - y = 17$ , find the value of  $(x - y)$ .

OR

(b) Sum of two numbers is 105 and their difference is 45. Find the numbers.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: Solve the following system of linear equations graphically:  $x - y + 1 = 0$ ;  $x + y = 5$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL\_SECTION: D

ORIGINAL\_MARKS: 5

QUESTION: (a) Using graphical method, solve the following system of equations:  $3x + y + 4 = 0$  and  $3x - y + 2 = 0$

OR

(b) Tara scored 40 marks in a test, getting 3 marks for each right answer and losing 1 mark for each wrong answer. Had 4 marks been awarded for each correct answer and 2 marks been deducted for each wrong answer, then Tara would have scored 50 marks. Assuming that Tara attempted all questions, find the total number of questions in the test.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL\_SECTION: D

ORIGINAL\_MARKS: 5

QUESTION: (a) Using graphical method, solve the following system of equations:  $3x + y + 4 = 0$  and  $3x - y + 2 = 0$

OR

(b) Tara scored 40 marks in a test, getting 3 marks for each right answer and losing 1 mark for each wrong answer. Had 4 marks been awarded for each correct answer and 2 marks been deducted for each wrong answer, then Tara would have scored 50 marks. Assuming that Tara attempted all questions, find the total number of questions in the test.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: In the given figure, graphs of two linear equations are shown. The pair of these linear equations is:

OPTIONS: (A) consistent with unique solution. (B) consistent with infinitely many solutions. (C) inconsistent. (D) inconsistent but can be made consistent by extending these lines.

ANSWER: (C) inconsistent.

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YEAR: 2024

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: Solve the following system of linear equations:  $2p + 3q = 13$  and  $5p - 4q = -2$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: Solve the following system of linear equations algebraically:  $2x + 5y = -4$ ;  $4x - 3y = 5$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: In a chemistry lab, there is some quantity of 50% acid solution and some quantity of 25% acid solution. How much of each should be mixed to make 10 litres of 40% acid solution?

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL\_SECTION: D

ORIGINAL\_MARKS: 5

QUESTION: (a) A train travels a distance of 90 km at a constant speed. Had the speed been 15 km/h more, it would have taken 30 minutes less for the journey. Find the original speed of the train.

OR

(b) Find the value of 'c' for which the quadratic equation  $(c+1)x^2 - 6(c+1)x + 3(c+9)=0$ ;  $c \neq -1$  has real and equal roots.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: A dealer sells an article for ₹75 and gains as much percent as the cost price of the article. Find the cost price of the article.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL\_SECTION: D

ORIGINAL\_MARKS: 5

QUESTION: (a) If three times the greater of two numbers is divided by the smaller one, we get 4 as the quotient and 3 as the remainder. Also, if seven times the smaller number is divided by greater one, we get 5 as the quotient and 1 as the remainder. Find the numbers.

OR

(b) Using graphical method, solve the following system of equations:  $3x - 2y = 10$  and  $5x + 3y = 4$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The pair of linear equations  $x + 2y + 5 = 0$  and  $-3x = 6y - 1$  has:

OPTIONS: (A) unique solution (B) exactly two solutions (C) infinitely many solutions (D) no solution

ANSWER: (D) no solution

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YEAR: 2024

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: (a) If  $2x + y = 13$  and  $4x - y = 17$ , find the value of  $(x - y)$ .

OR

(b) Sum of two numbers is 105 and their difference is 45. Find the numbers.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: Solve the following system of linear equations graphically:  $x - y + 1 = 0$ ;  $x + y = 5$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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**\*\*CHAPTER: 4 - QUADRATIC EQUATIONS\*\***

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YEAR: 2024

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The value of 'a' for which  $ax^2 + x + a = 0$  has equal and positive roots is:

OPTIONS: (A) 2 (B) -2 (C)  $\frac{1}{2}$  (D)  $-\frac{1}{2}$

ANSWER: (D)  $-\frac{1}{2}$

---

YEAR: 2024

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The discriminant of the quadratic equation  $b x^2 + a x + c = 0$ ;  $b \neq 0$  is given by:

OPTIONS: (A)  $b^2 - 4ac$  (B)  $\sqrt{b^2 - 4ac}$  (C)  $\sqrt{a^2 - 4bc}$  (D)  $a^2 - 4bc$

ANSWER: (D)  $a^2 - 4bc$

---

YEAR: 2024

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: Which of the following equations is a quadratic equation?

OPTIONS: (A)  $x^2 + 1 = (x - 1)^2$  (B)  $(x + \sqrt{x})^2 = 2x\sqrt{x}$  (C)  $x^3 + 3x^2 = (x + 1)^3$  (D)  $(x + 1)(x - 1) = (x + 1)^2$

ANSWER: (B)  $(x + \sqrt{x})^2 = 2x\sqrt{x}$

---

YEAR: 2024

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If  $x^2 + bx + b = 0$  has two real and distinct roots, then the value of  $b$  can be:

OPTIONS: (A) 0 (B) 4 (C) 3 (D) -3

ANSWER: (D) -3

---

YEAR: 2024

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The quadratic equation whose roots are 7 and  $\frac{1}{7}$  is:

OPTIONS: (A)  $7x^2 - 50x + 7 = 0$  (B)  $7x^2 - 50x + 1 = 0$  (C)  $7x^2 + 50x - 7 = 0$  (D)  $7x^2 + 50x - 1 = 0$

ANSWER: (A)  $7x^2 - 50x + 7 = 0$

---

YEAR: 2024

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The quadratic equation whose sum and product of roots are 'a' and '1/a' respectively is:

OPTIONS: (A)  $ax^2 - ax + 1 = 0$  (B)  $ax^2 - a^2x + 1 = 0$  (C)  $ax^2 + ax + 1 = 0$  (D)  $ax^2 + a^2x - 1 = 0$

ANSWER: (B)  $ax^2 - a^2x + 1 = 0$

---

YEAR: 2024

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The roots of the quadratic equation  $\sqrt{4x^2 - 5x + 4} = 0$  are:

OPTIONS: (A) irrational (B) rational and distinct (C) not real (D) rational and equal

ANSWER: (C) not real

---

YEAR: 2024

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: (a) Find the value of 'k' for which the quadratic equation  $4x^2 + kx + 1 = 0$  has real and equal roots.

OR

(b) If ' $\alpha$ ' and ' $\beta$ ' are the zeroes of the polynomial  $p(y) = y^2 - 5y + 3$ , then find the value of  $\alpha^4\beta^3 + \alpha^3\beta^4$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

---

YEAR: 2024

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: (a) If  $\sqrt{x = 2}$  is a solution of the quadratic equation  $\sqrt{kx^2 + 2x - 3} = 0$ , find the value of k.

OR

(b) Solve for  $\sqrt{x: \sqrt{2}x^2 + 7x + 5\sqrt{2}} = 0$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

---

YEAR: 2024

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL\_SECTION: D

ORIGINAL\_MARKS: 5

QUESTION: A 2-digit number is such that the product of its digits is 18. When 63 is subtracted from the number, the digits interchange their places. Find the number.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

---

YEAR: 2024

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: Three consecutive integers are such that sum of the square of second and product of other two is 161. Find the three integers.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

---

YEAR: 2024

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL\_SECTION: D

ORIGINAL\_MARKS: 5

QUESTION: (a) A train travels a distance of 90 km at a constant speed. Had the speed been 15 km/h more, it would have taken 30 minutes less for the journey. Find the original speed of the train.

OR

(b) Find the value of 'c' for which the quadratic equation  $(c+1)x^2 - 6(c+1)x + 3(c+9)=0$ ;  $c \neq -1$  has real and equal roots.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

---

YEAR: 2024

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: A rectangular floor area can be completely tiled with 200 square tiles. If the side length of each tile is increased by 1 unit, it would take only 128 tiles to cover the floor. (i) Assuming the original length of each side of a tile be  $x$  units, make a quadratic equation from the above information. (ii) Write the corresponding quadratic equation in standard form. (iii) (a) Find the value of  $x$ , the length of side of a tile by factorisation. OR (b) Solve the quadratic equation for  $x$ , using quadratic formula.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: Value of  $k$  for which  $(x = 2)$  is a solution of the equation  $(5x^2 - 4x + (2 + k) = 0)$  is:

OPTIONS: (A) 10 (B) -10 (C) 14 (D) -14

ANSWER: (B) -10

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YEAR: 2024

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If the discriminant of the quadratic equation  $(3x^2 - 2x + c = 0)$  is 16, then the value of  $c$  is:

OPTIONS: (A) 1 (B) 0 (C) -1 (D)  $\sqrt{2}$

ANSWER: (C) -1

---

YEAR: 2024

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL\_SECTION: D

ORIGINAL\_MARKS: 5

QUESTION: (a) Find the value of ' $k$ ' for which the quadratic equation  $((k+1)x^2 - 2(3k+1)x + (8k+1) = 0)$  has real and equal roots.

OR

(b) A 2-digit number is such that the product of its digits is 18. When 63 is subtracted from the number, the digits interchange their places. Find the number.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

---

YEAR: 2024

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If  $y = 1$  is one of the solutions of the quadratic equation  $py^2 + py + 3 = 0$  then the value of p is:

OPTIONS: (A) -3 (B) 2 (C)  $-\frac{3}{2}$  (D) -2

ANSWER: (C)  $-\frac{3}{2}$

---

YEAR: 2024

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: (a) A fraction becomes  $\frac{1}{3}$ , when 1 is subtracted from the numerator and it becomes  $\frac{1}{4}$ , when 8 is added to its denominator. Find the fraction.

OR

(b) The denominator of a fraction is one more than twice the numerator. If the sum of the fraction and its reciprocal is  $\frac{16}{21}$ , find the fraction.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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**\*\*CHAPTER: 5 - ARITHMETIC PROGRESSIONS\*\***

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YEAR: 2024

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The common difference of an A.P. in which  $a_{15} - a_{11} = 48$ , is:

OPTIONS: (A) 12 (B) 16 (C) -12 (D) -16

ANSWER: (A) 12

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YEAR: 2024

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The common difference of the A.P.  $\left(\frac{1}{2x}, \frac{1-4x}{2x}, \frac{1-8x}{2x}, \dots\right)$  is:

OPTIONS: (A)  $-2x$  (B)  $-2$  (C)  $2$  (D)  $2x$

ANSWER: (B)  $-2$

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YEAR: 2024

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: Three numbers in A.P. have the sum 30. What is its middle term?

OPTIONS: (A) 4 (B) 10 (C) 16 (D) 8

ANSWER: (B) 10

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YEAR: 2024

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: Which term of the A.P.  $-29, -26, -23, \dots, 61$  is 16?

OPTIONS: (A) 11th (B) 16th (C) 10th (D) 31st

ANSWER: (A) 11th

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YEAR: 2024

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The common difference of an A.P. in which  $a_{20} - a_{15} = 20$ , is:

OPTIONS: (A) 4 (B) 5 (C)  $4d$  (D)  $5d$

ANSWER: (A) 4

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YEAR: 2024

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The next (4th) term of the A.P.  $\sqrt{18}, \sqrt{50}, \sqrt{98}, \dots$  is:

OPTIONS: (A)  $\sqrt{128}$  (B)  $\sqrt{140}$  (C)  $\sqrt{162}$  (D)  $\sqrt{200}$   
ANSWER: (C)  $\sqrt{162}$

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YEAR: 2024  
CHAPTER: 5 - ARITHMETIC PROGRESSIONS  
ORIGINAL\_SECTION: A  
ORIGINAL\_MARKS: 1  
QUESTION: The sum of first 200 natural numbers is:  
OPTIONS: (A) 2010 (B) 2000 (C) 20100 (D) 21000  
ANSWER: (C) 20100

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YEAR: 2024  
CHAPTER: 5 - ARITHMETIC PROGRESSIONS  
ORIGINAL\_SECTION: A  
ORIGINAL\_MARKS: 1  
QUESTION: The number of terms in the A.P. 3, 6, 9, 12, ..., 111 is:  
OPTIONS: (A) 36 (B) 40 (C) 37 (D) 30  
ANSWER: (C) 37

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YEAR: 2024  
CHAPTER: 5 - ARITHMETIC PROGRESSIONS  
ORIGINAL\_SECTION: A  
ORIGINAL\_MARKS: 1  
QUESTION: If the first three terms of an A.P. are  $\sqrt{3p - 1}$ ,  $\sqrt{3p + 5}$ ,  $\sqrt{5p + 1}$  respectively;  
then the value of  $\sqrt{p}$  is:  
OPTIONS: (A) 2 (B) -3 (C) 4 (D) 5  
ANSWER: (D) 5

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YEAR: 2024  
CHAPTER: 5 - ARITHMETIC PROGRESSIONS  
ORIGINAL\_SECTION: C  
ORIGINAL\_MARKS: 3  
QUESTION: (a) If the sum of first  $m$  terms of an A.P. is same as sum of its first  $n$  terms ( $m \neq n$ ),  
then show that the sum of its first  $(m + n)$  terms is zero.  
OR

(b) In an A.P., the sum of three consecutive terms is 24 and the sum of their squares is 194.  
Find the numbers.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: (a) If the sum of first 7 terms of an A.P. is 49 and that of first 17 terms is 289, find the sum of its first 20 terms.

OR

(b) The ratio of the 10th term to its 30th term of an A.P. is 1:3 and the sum of its first six terms is 42. Find the first term and the common difference of A.P.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL\_SECTION: D

ORIGINAL\_MARKS: 5

QUESTION: (A) The sum of first and eighth terms of an A.P. is 32 and their product is 60. Find the first term and common difference of the A.P. Hence, also find the sum of its first 20 terms.

OR

(B) In an A.P. of 40 terms, the sum of first 9 terms is 153 and the sum of last 6 terms is 687. Determine the first term and common difference of A.P. Also, find the sum of all the terms of the A.P.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The sum of first  $n$  terms of an A.P. is  $\sqrt{3n^2 + 4n}$  and its common difference is 6, then its first term is:

OPTIONS: (A) 7 (B) 4 (C) 6 (D) 3

ANSWER: (A) 7

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YEAR: 2024

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: (a) If the sum of first 7 terms of an A.P. is 49 and that of first 17 terms is 289, find the sum of its first 20 terms.

OR

(b) The ratio of the 10th term to its 30th term of an A.P. is 1:3 and the sum of its first six terms is 42. Find the first term and the common difference of A.P.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The  $14^{\text{th}}$  term from the end of the A.P.  $(-11, -8, -5, \dots, 49)$  is:

OPTIONS: (A) 7 (B) 10 (C) 13 (D) 28

ANSWER: (B) 10

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YEAR: 2024

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL\_SECTION: D

ORIGINAL\_MARKS: 5

QUESTION: (a) The sum of the third term and the seventh term of an AP is 6 and their product is 8. Find the sum of the first sixteen terms of the AP.

OR

(b) The minimum age of children eligible to participate in a painting competition is 8 years. It is observed that the age of the youngest boy was 8 years and the ages of the participants, when seated in order of age, have a common difference of 4 months. If the sum of the ages of all the participants is 168 years, find the age of the eldest participant in the painting competition.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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**\*\*CHAPTER: 6 - COORDINATE GEOMETRY\*\***

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YEAR: 2024

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The distance of a point A from x-axis is 3 units. Which of the following cannot be coordinates of the point A?

OPTIONS: (A) (1, 3) (B) (-3, -3) (C) (-3, 3) (D) (3, 1)

ANSWER: (D) (3, 1)

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YEAR: 2024

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The coordinates of the end points of a diameter of a circle are (5, -2) and (5, 2). The length of the radius of the circle is:

OPTIONS: (A)  $\pm 2$  (B)  $\pm 4$  (C) 4 (D) 2

ANSWER: (D) 2

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YEAR: 2024

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The distance between the points  $(a \cos \theta, -a \sin \theta)$  and  $(a \sin \theta, a \cos \theta)$  is:

OPTIONS: (A) a (B)  $a\sqrt{2}$  (C) 0 (D) 2a

ANSWER: (B)  $a\sqrt{2}$

---

YEAR: 2024

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The mid-point of the line segment joining the points P(-4, 5) and Q(4, 6) lies on:

OPTIONS: (A) x-axis (B) y-axis (C) origin (D) neither x-axis nor y-axis

ANSWER: (B) y-axis

---

YEAR: 2024

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The points  $(-5, 0)$ ,  $(5, 0)$  and  $(0, 4)$  are the vertices of a triangle which is a/an:

OPTIONS: (A) right-angled triangle (B) isosceles triangle (C) equilateral triangle (D) scalene triangle

ANSWER: (B) isosceles triangle

---

YEAR: 2024

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The line represented by the equation  $x - y = 0$  is:

OPTIONS: (A) parallel to x-axis (B) parallel to y-axis (C) passing through the origin (D) passing through the point  $(3, 2)$

ANSWER: (C) passing through the origin

---

YEAR: 2024

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: AOBC is a rectangle whose three vertices are  $A(0, 2)$ ,  $O(0, 0)$  and  $B(4, 0)$ . The square of the length of its diagonal is equal to:

OPTIONS: (A) 36 (B) 20 (C) 16 (D) 4

ANSWER: (B) 20

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YEAR: 2024

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If the mid-point of the line segment joining the points  $(a, 4)$  and  $(2, 2b)$  is  $(2, 6)$ , then the value of  $(a + b)$  is given by:

OPTIONS: (A) 6 (B) 7 (C) 8 (D) 16

ANSWER: (A) 6

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YEAR: 2024

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: Point P divides the line segment joining the points A(4, -5) and B(1, 2) in the ratio 5:2. Co-ordinates of point P are:

OPTIONS: (A)  $(\frac{5}{2}, -\frac{3}{2})$  (B)  $(\frac{11}{7}, 0)$  (C)  $(\frac{13}{7}, 0)$  (D)  $(0, \frac{13}{7})$

ANSWER: (C)  $(\frac{13}{7}, 0)$

---

YEAR: 2024

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: XOYZ is a rectangle with vertices X(-3, 0), O(0, 0), Y(0, 4) and Z(x, y). The length of its each diagonal is:

OPTIONS: (A) 5 units (B)  $\sqrt{5}$  units (C)  $x^2 + y^2$  units (D) 4 units

ANSWER: (A) 5 units

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YEAR: 2024

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: Find the ratio in which the point P(-4, 6) divides the line segment joining the points A(-6, 10) and B(3, -8).

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

---

YEAR: 2024

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: Prove that the points (3, 0), (6, 4) and (-1, 3) are the vertices of an isosceles triangle.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: (a) Find the ratio in which the line segment joining the points (3, -5) and (-1, 6) is divided by the line  $y = x$ .

OR

(b) A(3, 0), B(6, 4) and C(-1, 3) are vertices of a triangle ABC. Find length of its median BE.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

---

YEAR: 2024

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: (a) Find the ratio in which the line segment joining the points (5, 3) and (-1, 6) is divided by Y-axis.

OR

(b) P(-2, 5) and Q(3, 2) are two points. Find the coordinates of the point R on line segment PQ such that  $PR = 2QR$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

---

YEAR: 2024

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The fourth vertex D of a parallelogram ABCD whose three vertices are A(-2, 3), B(6, 7) and C(8, 3) is:

OPTIONS: (A) (0, 1) (B) (0, -1) (C) (-1, 0) (D) (1, 0)

ANSWER: (B) (0, -1)

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YEAR: 2024

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: (a) Find the ratio in which the line segment joining the points (5, 3) and (-1, 6) is divided by Y-axis.

OR

(b) P(-2, 5) and Q(3, 2) are two points. Find the coordinates of the point R on line segment PQ such that  $PR = 2QR$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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**\*\*CHAPTER: 7 - TRIANGLES\*\***

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YEAR: 2024

CHAPTER: 7 - TRIANGLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: In triangles ABC and DEF,  $\angle B = \angle E$ ,  $\angle F = \angle C$  and  $AB = 3 DE$ . Then, the two triangles are:

OPTIONS: (A) congruent but not similar (B) congruent as well as similar (C) neither congruent nor similar (D) similar but not congruent

ANSWER: (D) similar but not congruent

---

YEAR: 2024

CHAPTER: 7 - TRIANGLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: In the given figure,  $PQ \parallel BC$ . If  $AP/PB = 4/13$  and  $AC = 20.4$  cm, then the length of AQ is:

OPTIONS: (A) 2.8 cm (B) 5.8 cm (C) 3.8 cm (D) 4.8 cm

ANSWER: (D) 4.8 cm

---

YEAR: 2024

CHAPTER: 7 - TRIANGLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: Which of the following statements is incorrect?

OPTIONS: (A) Two congruent figures are always similar. (B) A square and a rhombus of the same area are always similar. (C) Two equilateral triangles are always similar. (D) Two similar triangles need not be congruent.

ANSWER: (B) A square and a rhombus of the same area are always similar.

---

YEAR: 2024

CHAPTER: 7 - TRIANGLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: In the given figure, PQ is a tangent to a circle with centre O. If  $\angle NML = 30^\circ$ , then  $\angle RLM$  is:

OPTIONS: (A)  $30^\circ$  (B)  $60^\circ$  (C)  $90^\circ$  (D)  $120^\circ$

ANSWER: (B)  $60^\circ$

---

YEAR: 2024

CHAPTER: 7 - TRIANGLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: E and F are points on the sides AB and AC respectively of a  $\triangle ABC$  such that  $AE/EB = AF/FC = 1/2$ . Which of the following relation is true?

OPTIONS: (A)  $EF = 2BC$  (B)  $BC = 2EF$  (C)  $EF = 3BC$  (D)  $BC = 3EF$

ANSWER: (D)  $BC = 3EF$

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YEAR: 2024

CHAPTER: 7 - TRIANGLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: In the given figure, in  $\triangle ABC$ ,  $DE \parallel BC$ . If  $AD = 4$  cm,  $AB = 9$  cm and  $AC = 13.5$  cm, then the length of EC is:

OPTIONS: (A) 6 cm (B) 7.5 cm (C) 9 cm (D) 5.7 cm

ANSWER: (B) 7.5 cm

---

YEAR: 2024

CHAPTER: 7 - TRIANGLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: In the given figure, in  $\triangle ABC$ ,  $DE \parallel BC$ . If  $AD = 2$  cm,  $BD = 3$  cm,  $BC = 7.5$  cm, then the length of  $DE$  (in cm) is:

OPTIONS: (A) 2.5 (B) 3 (C) 5 (D) 6

ANSWER: (B) 3

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YEAR: 2024

CHAPTER: 7 - TRIANGLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: In the given figure, in  $\triangle ABC$ ,  $DE \parallel BC$ . If  $AD = 2.4$  cm,  $DB = 4$  cm and  $AE = 2$  cm, then the length of  $AC$  is:

OPTIONS: (A)  $10/3$  cm (B)  $3/10$  cm (C)  $16/3$  cm (D) 1.2 cm

ANSWER: (A)  $10/3$  cm

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YEAR: 2024

CHAPTER: 7 - TRIANGLES

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: In the given figure,  $\frac{EA}{EC} = \frac{EB}{ED}$ , prove that  $\triangle EAB \sim \triangle ECD$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 7 - TRIANGLES

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: In the given figure,  $\frac{QR}{QS} = \frac{QT}{PR}$  and  $\angle 1 = \angle 2$ , show that  $\triangle PQS \sim \triangle TQR$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 7 - TRIANGLES

ORIGINAL\_SECTION: D

ORIGINAL\_MARKS: 5

QUESTION: (A) If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then prove that the other two sides are divided in the same ratio.

OR

(B) In the given figure PA, QB and RC are each perpendicular to AC. If  $AP = x$ ,  $BQ = y$  and  $CR = z$ , then prove that  $1/x + 1/z = 1/y$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 7 - TRIANGLES

ORIGINAL\_SECTION: D

ORIGINAL\_MARKS: 5

QUESTION: (A) If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then prove that the other two sides are divided in the same ratio.

OR

(B) In the given figure PA, QB and RC are each perpendicular to AC. If  $AP = x$ ,  $BQ = y$  and  $CR = z$ , then prove that  $1/x + 1/z = 1/y$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 7 - TRIANGLES

ORIGINAL\_SECTION: D

ORIGINAL\_MARKS: 5

QUESTION: (A) If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then prove that the other two sides are divided in the same ratio.

OR

(B) Sides AB and AC and median AD of  $\triangle ABC$  are respectively proportional to sides PQ and PR and median PM of another  $\triangle PQR$ . Show that  $\triangle ABC \sim \triangle PQR$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

---

YEAR: 2024

CHAPTER: 7 - TRIANGLES

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: In the given figure,  $\triangle AHK \sim \triangle ABC$ . If  $AK = 8$  cm,  $BC = 3.2$  cm and  $HK = 6.4$  cm, then find the length of AC.

OPTIONS: Not an MCQ.  
ANSWER: Not applicable.

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YEAR: 2024  
CHAPTER: 7 - TRIANGLES  
ORIGINAL\_SECTION: C  
ORIGINAL\_MARKS: 3  
QUESTION: Prove that the tangents drawn at the end points of a chord of a circle makes equal angles with the chord.  
OPTIONS: Not an MCQ.  
ANSWER: Not applicable.

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\*\*CHAPTER: 8 - CIRCLES\*\*  
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YEAR: 2024  
CHAPTER: 8 - CIRCLES  
ORIGINAL\_SECTION: A  
ORIGINAL\_MARKS: 1  
QUESTION: The tangents drawn at the extremities of the diameter of a circle are always:  
OPTIONS: (A) parallel (B) perpendicular (C) equal (D) intersecting  
ANSWER: (A) parallel

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YEAR: 2024  
CHAPTER: 8 - CIRCLES  
ORIGINAL\_SECTION: A  
ORIGINAL\_MARKS: 1  
QUESTION: In the given figure, PA is a tangent from an external point P to a circle with centre O. If  $\angle POB = 115^\circ$ , then  $\angle APO$  is equal to:  
OPTIONS: (A)  $25^\circ$  (B)  $65^\circ$  (C)  $90^\circ$  (D)  $35^\circ$   
ANSWER: (A)  $25^\circ$

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YEAR: 2024  
CHAPTER: 8 - CIRCLES  
ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If the length of a chord of a circle is equal to its radius, then the angle subtended by chord at the centre is:

OPTIONS: (A)  $60^\circ$  (B)  $30^\circ$  (C)  $120^\circ$  (D)  $90^\circ$

ANSWER: (A)  $60^\circ$

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YEAR: 2024

CHAPTER: 8 - CIRCLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: In the given figure, AB is the chord of the larger circle touching the smaller circle. The centre of both the circles is O. If  $AB = 2r$  and  $OP = r$ , then the radius of larger circle is:

OPTIONS: (A)  $2r$  (B)  $3r$  (C)  $2\sqrt{2}r$  (D)  $\sqrt{2}r$

ANSWER: (D)  $\sqrt{2}r$

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YEAR: 2024

CHAPTER: 8 - CIRCLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: A parallelogram having one of its sides 5 cm circumscribes a circle. The perimeter of parallelogram is:

OPTIONS: (A) 20 cm (B) less than 20 cm (C) more than 20 cm but less than 40 cm (D) 40 cm

ANSWER: (A) 20 cm

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YEAR: 2024

CHAPTER: 8 - CIRCLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: In the given figure, PA and PB are tangents to a circle with centre O such that  $\angle P = 90^\circ$ . If  $AB = 3\sqrt{2}$  cm, then the diameter of the circle is:

OPTIONS: (A)  $3\sqrt{2}$  cm (B)  $6\sqrt{2}$  cm (C) 3 cm (D) 6 cm

ANSWER: (D) 6 cm

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YEAR: 2024

CHAPTER: 8 - CIRCLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: In the given figure, if PT is a tangent to a circle with centre O and  $\angle TPO = 35^\circ$ , then the measure of  $\angle x$  is:

OPTIONS: (A)  $110^\circ$  (B)  $115^\circ$  (C)  $120^\circ$  (D)  $125^\circ$

ANSWER: (D)  $125^\circ$

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YEAR: 2024

CHAPTER: 8 - CIRCLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: In the given figure, O is the centre of the circle. MN is the chord and the tangent ML at point M makes an angle of  $70^\circ$  with MN. The measure of  $\angle MON$  is:

OPTIONS: (A)  $120^\circ$  (B)  $140^\circ$  (C)  $70^\circ$  (D)  $90^\circ$

ANSWER: (B)  $140^\circ$

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YEAR: 2024

CHAPTER: 8 - CIRCLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: Maximum number of common tangents that can be drawn to two circles intersecting at two distinct points is:

OPTIONS: (A) 4 (B) 3 (C) 2 (D) 1

ANSWER: (C) 2

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YEAR: 2024

CHAPTER: 8 - CIRCLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: In the given figure, RJ and RL are two tangents to the circle. If  $\angle RJL = 42^\circ$ , then the measure of  $\angle JOL$  is:

OPTIONS: (A)  $42^\circ$  (B)  $84^\circ$  (C)  $96^\circ$  (D)  $138^\circ$

ANSWER: (D)  $138^\circ$

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YEAR: 2024

CHAPTER: 8 - CIRCLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: In the given figure, AB and AC are tangents to the circle. If  $\angle ABC = 42^\circ$ , then the measure of  $\angle BAC$  is:

OPTIONS: (A)  $96^\circ$  (B)  $42^\circ$  (C)  $106^\circ$  (D)  $86^\circ$

ANSWER: (A)  $96^\circ$

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YEAR: 2024

CHAPTER: 8 - CIRCLES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: In the given figure, QR is a common tangent to the two given circles touching externally at A. The tangent at A meets QR at P. If  $AP = 4.2$  cm, then the length of QR is:

OPTIONS: (A) 4.2 cm (B) 2.1 cm (C) 8.4 cm (D) 6.3 cm

ANSWER: (C) 8.4 cm

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YEAR: 2024

CHAPTER: 8 - CIRCLES

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: (a) If two tangents inclined at an angle of  $60^\circ$  are drawn to a circle of radius 3 cm, then find the length of each tangent.

OR

(b) Prove that the tangents drawn at the ends of a diameter of a circle are parallel.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 8 - CIRCLES

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: In the given figure, AB and CD are tangents to a circle centred at O. Is  $\angle BAC = \angle DCA$ ? Justify your answer.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 8 - CIRCLES

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: (a) In the given figure, PQ is tangent to a circle centred at O and  $\angle BAQ = 30^\circ$ ; show that  $BP = BQ$ .

OR

(b) In the given figure, AB, BC, CD and DA are tangents to the circle with centre O forming a quadrilateral ABCD. Show that  $\angle AOB + \angle COD = 180^\circ$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 8 - CIRCLES

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: Prove that the tangents drawn at the end points of a chord of a circle makes equal angles with the chord.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 8 - CIRCLES

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: From an external point P, two tangents PA and PB are drawn to the circle with centre O. Prove that OP is the perpendicular bisector of chord AB.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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**\*\*CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY & TRIGONOMETRIC IDENTITIES\*\***

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YEAR: 2024

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY & TRIGONOMETRIC IDENTITIES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION:  $\tan 2A = 3 \tan A$  is true, when the measure of  $\angle A$  is:

OPTIONS: (A)  $90^\circ$  (B)  $60^\circ$  (C)  $45^\circ$  (D)  $30^\circ$

ANSWER: (D)  $30^\circ$

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YEAR: 2024

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY & TRIGONOMETRIC IDENTITIES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: Which of the following statements is true?

OPTIONS: (A)  $\sin 20^\circ > \sin 70^\circ$  (B)  $\sin 20^\circ > \cos 20^\circ$  (C)  $\cos 20^\circ > \cos 70^\circ$  (D)  $\tan 20^\circ > \tan 70^\circ$

ANSWER: (C)  $\cos 20^\circ > \cos 70^\circ$

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YEAR: 2024

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY & TRIGONOMETRIC IDENTITIES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If  $\sin \theta = \cos \theta$  ( $0^\circ < \theta < 90^\circ$ ), then value of  $(\sec \theta \cdot \sin \theta)$  is:

OPTIONS: (A)  $1/\sqrt{2}$  (B)  $\sqrt{2}$  (C) 1 (D) 0

ANSWER: (B)  $\sqrt{2}$

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YEAR: 2024

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY & TRIGONOMETRIC IDENTITIES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If  $\cos(\alpha + \beta) = 0$ , then value of  $\cos((\alpha + \beta)/2)$  is equal to:

OPTIONS: (A)  $1/\sqrt{2}$  (B)  $1/2$  (C) 0 (D)  $\sqrt{2}$

ANSWER: (A)  $1/\sqrt{2}$

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YEAR: 2024

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY & TRIGONOMETRIC IDENTITIES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If  $\sec \theta - \tan \theta = m$ , then the value of  $\sec \theta + \tan \theta$  is:

OPTIONS: (A)  $1 - 1/m$  (B)  $m^2 - 1$  (C)  $1/m$  (D)  $-m$

ANSWER: (C)  $1/m$

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YEAR: 2024

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY & TRIGONOMETRIC IDENTITIES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If  $x/3 = 2 \sin A$ ,  $y/3 = 2 \cos A$ , then the value of  $x^2 + y^2$  is:

OPTIONS: (A) 36 (B) 9 (C) 6 (D) 18

ANSWER: (A) 36

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YEAR: 2024

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY & TRIGONOMETRIC IDENTITIES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If  $4 \sec \theta - 5 = 0$ , then the value of  $\cot \theta$  is:

OPTIONS: (A)  $3/4$  (B)  $4/5$  (C)  $5/3$  (D)  $4/3$

ANSWER: (D)  $4/3$

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YEAR: 2024

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY & TRIGONOMETRIC IDENTITIES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If  $\sin \theta = 1$ , then the value of  $(1/2) \sin (\theta/2)$  is:

OPTIONS: (A)  $1/(2\sqrt{2})$  (B)  $1/\sqrt{2}$  (C)  $1/2$  (D) 0

ANSWER: (A)  $1/(2\sqrt{2})$

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YEAR: 2024

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY & TRIGONOMETRIC IDENTITIES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The value of  $\tan^2 \theta - (1/\cos \theta \times \sec \theta)$  is:

OPTIONS: (A) 1 (B) 0 (C) -1 (D) 2

ANSWER: (C) -1

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YEAR: 2024

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY & TRIGONOMETRIC IDENTITIES

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: (a) If  $x \cos 60^\circ + y \cos 0^\circ + \sin 30^\circ - \cot 45^\circ = 5$ , then find the value of  $x + 2y$ .

OR

(b) Evaluate:  $\tan^2 60^\circ / (\sin^2 60^\circ + \cos^2 30^\circ)$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY & TRIGONOMETRIC IDENTITIES

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: Prove that:  $\tan\theta/(1 - \cot\theta) + \cot\theta/(1 - \tan\theta) = 1 + \sec\theta \operatorname{cosec}\theta$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY & TRIGONOMETRIC IDENTITIES

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: (a) Evaluate:  $2 \sin^2 30^\circ \sec 60^\circ + \tan^2 60^\circ$ .

OR

(b) If  $2 \sin(A + B) = \sqrt{3}$  and  $\cos(A - B) = 1$ , then find the measures of angles A and B,  $0 \leq A, B$ ,  $(A + B) \leq 90^\circ$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY & TRIGONOMETRIC IDENTITIES

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: Prove that:  $(\sin\theta - 2\sin^3\theta) / (2\cos^3\theta - \cos\theta) = \tan\theta$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY & TRIGONOMETRIC IDENTITIES

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: Prove that:  $(\sin\theta + \cos\theta - 1) / (\sin\theta + \cos\theta - 1) = 1/(\sec\theta - \tan\theta)$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY & TRIGONOMETRIC IDENTITIES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If  $\theta$  is an acute angle and  $7 + 4 \sin \theta = 9$ , then the value of  $\theta$  is:

OPTIONS: (A)  $90^\circ$  (B)  $30^\circ$  (C)  $45^\circ$  (D)  $60^\circ$

ANSWER: (B)  $30^\circ$

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YEAR: 2024

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY & TRIGONOMETRIC IDENTITIES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If  $\cos \theta = \sqrt{3}/2$  and  $\sin \phi = 1/2$ , then  $\tan (\theta + \phi)$  is:

OPTIONS: (A)  $\sqrt{3}$  (B)  $1/\sqrt{3}$  (C) 1 (D) not defined

ANSWER: (C) 1

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YEAR: 2024

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY & TRIGONOMETRIC IDENTITIES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If  $\sin \alpha = \sqrt{3}/2$ ,  $\cos \beta = \sqrt{3}/2$ , then  $\tan \alpha \cdot \tan \beta$  is:

OPTIONS: (A)  $\sqrt{3}$  (B)  $1/\sqrt{3}$  (C) 1 (D) 0

ANSWER: (C) 1

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YEAR: 2024

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY & TRIGONOMETRIC IDENTITIES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The value of  $\theta$  for which  $\sqrt{2}\sin^2\theta = \frac{1}{2}$  ;  $(0^\circ \leq \theta \leq 90^\circ)$  is:

OPTIONS: (A)  $30^\circ$  (B)  $60^\circ$  (C)  $45^\circ$  (D)  $90^\circ$

ANSWER: (A)  $30^\circ$

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YEAR: 2024

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY & TRIGONOMETRIC IDENTITIES

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: (a) If  $\sin(A - B) = 1/2$ ,  $\cos(A + B) = 1/2$  ;  $0 < A + B \leq 90^\circ$ ,  $A > B$ ; find  $\angle A$  and  $\angle B$ .  
OR

(b) Evaluate:  $(5 \cos^2 60^\circ + 4 \sec^2 30^\circ - \tan^2 45^\circ) / (\sin^2 30^\circ + \sin^2 60^\circ)$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY & TRIGONOMETRIC IDENTITIES

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: Prove that:  $\sqrt{\sec^2\theta + \operatorname{cosec}^2\theta} = \tan\theta + \cot\theta$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY & TRIGONOMETRIC IDENTITIES

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: Prove that:  $\frac{\sin\theta}{1 + \cos\theta} + \frac{1 + \cos\theta}{\sin\theta} = 2 \operatorname{cosec}\theta$ .

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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\*\*CHAPTER: 10 - HEIGHTS AND DISTANCES\*\*  
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YEAR: 2024

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: A 30 m long rope is tightly stretched and tied from the top of pole to the ground. If the rope makes an angle of  $60^\circ$  with the ground, the height of the pole is:

OPTIONS: (A)  $10\sqrt{3}$  m (B)  $30\sqrt{3}$  m (C) 15 m (D)  $15\sqrt{3}$  m

ANSWER: (D)  $15\sqrt{3}$  m

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YEAR: 2024

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: A kite is flying at a height of 150 m from the ground. It is attached to a string inclined at an angle of  $30^\circ$  to the horizontal. The length of the string is:

OPTIONS: (A)  $100\sqrt{3}$  m (B) 300 m (C)  $150\sqrt{2}$  m (D)  $150\sqrt{3}$  m

ANSWER: (B) 300 m

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YEAR: 2024

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: A ladder 14 m long leans against a wall. If the foot of the ladder is 7 m from the wall, then the angle of elevation of the top of the wall is:

OPTIONS: (A)  $15^\circ$  (B)  $30^\circ$  (C)  $45^\circ$  (D)  $60^\circ$

ANSWER: (D)  $60^\circ$

---

YEAR: 2024

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: From a point on the ground, which is 30 m away from the foot of a vertical tower, the angle of elevation of the top of the tower is found to be  $60^\circ$ . The height (in metres) of the tower is:

OPTIONS: (A)  $10\sqrt{3}$  (B)  $30\sqrt{3}$  (C) 60 (D) 30

ANSWER: (B)  $30\sqrt{3}$

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YEAR: 2024

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: At some time of the day, the length of the shadow of a tower is equal to its height. Then, the Sun's altitude at that time is:

OPTIONS: (A)  $30^\circ$  (B)  $45^\circ$  (C)  $60^\circ$  (D)  $90^\circ$

ANSWER: (B)  $45^\circ$

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YEAR: 2024

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The ratio of the length of a pole and its shadow on the ground is  $1:\sqrt{3}$ . The angle of elevation of the Sun is:

OPTIONS: (A)  $90^\circ$  (B)  $60^\circ$  (C)  $45^\circ$  (D)  $30^\circ$

ANSWER: (D)  $30^\circ$

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YEAR: 2024

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The length of the shadow of a tower on the plane ground is  $\sqrt{3}$  times the height of the tower. The angle of elevation of the Sun is:

OPTIONS: (A)  $30^\circ$  (B)  $45^\circ$  (C)  $60^\circ$  (D)  $90^\circ$

ANSWER: (A)  $30^\circ$

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YEAR: 2024

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL\_SECTION: D

ORIGINAL\_MARKS: 5

QUESTION: From the top of a 45 m high light house, the angles of depression of two ships, on the opposite side of it, are observed to be  $30^\circ$  and  $60^\circ$ . If the line joining the ships passes through the foot of the light house, find the distance between the ships. (Use  $\sqrt{3} = 1.73$ )

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL\_SECTION: D

ORIGINAL\_MARKS: 5

QUESTION: A person standing on the bank of a river observes that the angle of elevation of the top of a tower on the opposite bank is  $60^\circ$ . When he moves 30 m away from the bank, he finds the angle of elevation to be  $30^\circ$ . Find the height of the tower and width of the river. (Take  $\sqrt{3} = 1.732$ )

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL\_SECTION: D

ORIGINAL\_MARKS: 5

QUESTION: The angle of elevation of a jet plane from a point A on the ground is  $60^\circ$ . After a flight of 30 seconds, the angle of elevation changes to  $30^\circ$ . If the jet plane is flying at a constant height of  $3600\sqrt{3}$  m, find the speed of the jet plane.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL\_SECTION: D

ORIGINAL\_MARKS: 5

QUESTION: From the top of a 15 m high building, the angle of elevation of the top of a tower is found to be  $30^\circ$ . From the bottom of the same building, the angle of elevation of the top of the tower is found to be  $60^\circ$ . Find the height of the tower and the distance between tower and the building.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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**\*\*CHAPTER: 11 - MENSURATION\*\***

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YEAR: 2024

CHAPTER: 11 - MENSURATION

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: A solid sphere is cut into two hemispheres. The ratio of the surface areas of sphere to that of two hemispheres taken together, is:

OPTIONS: (A) 1:1 (B) 1:4 (C) 2:3 (D) 3:2

ANSWER: (C) 2:3

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YEAR: 2024

CHAPTER: 11 - MENSURATION

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The volume of the largest right circular cone that can be carved out from a solid cube of edge 2 cm is:

OPTIONS: (A)  $4\pi/3$  cu cm (B)  $5\pi/3$  cu cm (C)  $8\pi/3$  cu cm (D)  $2\pi/3$  cu cm

ANSWER: (D)  $2\pi/3$  cu cm

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YEAR: 2024

CHAPTER: 11 - MENSURATION

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: On the top face of the wooden cube of side 7 cm, hemispherical depressions of radius 0.35 cm are to be formed by taking out the wood. The maximum number of depressions that can be formed is:

OPTIONS: (A) 400 (B) 100 (C) 20 (D) 10

ANSWER: (B) 100

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YEAR: 2024

CHAPTER: 11 - MENSURATION

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If the area of a sector of a circle is  $7/20$  of the area of the circle, then the angle at the centre is equal to:

OPTIONS: (A)  $110^\circ$  (B)  $130^\circ$  (C)  $100^\circ$  (D)  $126^\circ$

ANSWER: (D)  $126^\circ$

---

YEAR: 2024

CHAPTER: 11 - MENSURATION

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If an arc subtends an angle of  $90^\circ$  at the centre of a circle, then the ratio of its length to the circumference of the circle is:

OPTIONS: (A) 2:3 (B) 1:4 (C) 4:1 (D) 1:3

ANSWER: (B) 1:4

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YEAR: 2024

CHAPTER: 11 - MENSURATION

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The area of the sector of a circle of radius 12 cm is  $60\pi \text{ cm}^2$ . The central angle of this sector is:

OPTIONS: (A)  $120^\circ$  (B)  $6^\circ$  (C)  $75^\circ$  (D)  $150^\circ$

ANSWER: (D)  $150^\circ$

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YEAR: 2024

CHAPTER: 11 - MENSURATION

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The ratio of total surface area of a solid hemisphere to the square of its radius is:

OPTIONS: (A)  $2\pi:1$  (B)  $4\pi:1$  (C)  $3\pi:1$  (D)  $1:4\pi$

ANSWER: (C)  $3\pi:1$

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YEAR: 2024

CHAPTER: 11 - MENSURATION

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The perimeter of the sector of a circle of radius 21 cm which subtends an angle of  $60^\circ$  at the centre of circle, is:

OPTIONS: (A) 22 cm (B) 43 cm (C) 64 cm (D) 462 cm

ANSWER: (C) 64 cm

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YEAR: 2024

CHAPTER: 11 - MENSURATION

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The length of an arc of a circle with radius 12 cm is  $10\pi$  cm. The angle subtended by the arc at the centre of the circle, is:

OPTIONS: (A)  $120^\circ$  (B)  $6^\circ$  (C)  $75^\circ$  (D)  $150^\circ$

ANSWER: (D)  $150^\circ$

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YEAR: 2024

CHAPTER: 11 - MENSURATION

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: A chord of a circle of radius 10 cm subtends a right angle at its centre. The length of the chord (in cm) is:

OPTIONS: (A)  $5\sqrt{2}$  (B)  $10\sqrt{2}$  (C)  $5/\sqrt{2}$  (D) 5

ANSWER: (B)  $10\sqrt{2}$

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YEAR: 2024

CHAPTER: 11 - MENSURATION

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The area of the square inscribed in a circle of radius  $5\sqrt{2}$  cm is:

OPTIONS: (A)  $50 \text{ cm}^2$  (B)  $100 \text{ cm}^2$  (C)  $25 \text{ cm}^2$  (D)  $200 \text{ cm}^2$

ANSWER: (B)  $100 \text{ cm}^2$

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YEAR: 2024

CHAPTER: 11 - MENSURATION

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: Two identical solid cubes of side 'a' are joined end-to-end. The total surface area of the resulting cuboid is:

OPTIONS: (A)  $6a^2$  (B)  $10a^2$  (C)  $5a^2$  (D)  $4a^2$

ANSWER: (B)  $10a^2$

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YEAR: 2024

CHAPTER: 11 - MENSURATION

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: (a) The minute hand of a clock is 14 cm long. Find the area on the face of the clock described by the minute hand in 5 minutes.

OR

(b) Find the length of the arc of a circle which subtends an angle of  $60^\circ$  at the centre of the circle of radius 42 cm.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 11 - MENSURATION

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: The inner and outer radii of a hollow cylinder surmounted on a hollow hemisphere of same radii are 3 cm and 4 cm respectively. If height of the cylinder is 14 cm, then find its total surface area (inner and outer).

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 11 - MENSURATION

ORIGINAL\_SECTION: D

ORIGINAL\_MARKS: 5

QUESTION: A vessel is in the form of a hollow hemisphere mounted by a hollow cylinder. The diameter of the hemisphere is 14 cm and the total height of the vessel is 13 cm. Find the inner surface area and the volume of the vessel.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 11 - MENSURATION

ORIGINAL\_SECTION: D

ORIGINAL\_MARKS: 5

QUESTION: A solid iron pole consists of a solid cylinder of height 200 cm and base diameter 28 cm, which is surmounted by another cylinder of height 50 cm and radius 7 cm. Find the mass of the pole, given that  $1 \text{ cm}^3$  of iron has approximately 8 g mass.

OPTIONS: Not an MCQ.  
ANSWER: Not applicable.

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YEAR: 2024  
CHAPTER: 11 - MENSURATION  
ORIGINAL\_SECTION: D  
ORIGINAL\_MARKS: 5

QUESTION: A medicine capsule is in the shape of a cylinder with two hemispheres stuck to each of its ends. The length of the entire capsule is 14 mm and the diameter of the capsule is 4 mm, find its surface area. Also, find its volume.

OPTIONS: Not an MCQ.  
ANSWER: Not applicable.

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YEAR: 2024  
CHAPTER: 11 - MENSURATION  
ORIGINAL\_SECTION: C  
ORIGINAL\_MARKS: 3

QUESTION: 14 cm long a hollow right circular cylinder has the difference between outer and inner radii as 1 cm. If the volume of the metal used in making the cylinder is  $176 \text{ cm}^3$ , find the outer and inner radii of the cylinder.

OPTIONS: Not an MCQ.  
ANSWER: Not applicable.

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YEAR: 2024  
CHAPTER: 11 - MENSURATION  
ORIGINAL\_SECTION: A  
ORIGINAL\_MARKS: 1

QUESTION: The perimeter of a sector of a circle whose central angle is  $90^\circ$  and radius 7 cm is:

OPTIONS: (A) 35 cm (B) 11 cm (C) 22 cm (D) 25 cm  
ANSWER: (D) 25 cm

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\*\*CHAPTER: 12 - STATISTICS\*\*  
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YEAR: 2024

CHAPTER: 12 - STATISTICS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The cumulative frequency for calculating median is obtained by adding the frequencies of all the:

OPTIONS: (A) classes up to the median class (B) classes following the median class (C) classes preceding the median class (D) all classes

ANSWER: (C) classes preceding the median class

---

YEAR: 2024

CHAPTER: 12 - STATISTICS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If mean and median of given set of observations are 10 and 11 respectively, then the value of mode is:

OPTIONS: (A) 10.5 (B) 8 (C) 13 (D) 21

ANSWER: (C) 13

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YEAR: 2024

CHAPTER: 12 - STATISTICS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: Mode and Mean of a data are  $15x$  and  $18x$ , respectively. Then the median of the data is:

OPTIONS: (A)  $x$  (B)  $11x$  (C)  $17x$  (D)  $34x$

ANSWER: (C)  $17x$

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YEAR: 2024

CHAPTER: 12 - STATISTICS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If the mean of 6, 7,  $p$ , 8,  $q$ , 14 is 9, then:

OPTIONS: (A)  $p - q = 19$  (B)  $p + q = 19$  (C)  $p - q = 21$  (D)  $p + q = 21$

ANSWER: (B)  $p + q = 19$

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YEAR: 2024

CHAPTER: 12 - STATISTICS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If the difference of mode and median of a data is 24, then the difference of its median and mean is:

OPTIONS: (A) 12 (B) 24 (C) 8 (D) 36

ANSWER: (A) 12

---

YEAR: 2024

CHAPTER: 12 - STATISTICS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The mean of five observations is 15. If the mean of first three observations is 14 and that of the last three observations is 17, then the third observation is:

OPTIONS: (A) 20 (B) 19 (C) 18 (D) 17

ANSWER: (C) 18

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YEAR: 2024

CHAPTER: 12 - STATISTICS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If the mean of the first  $n$  natural numbers is  $\frac{5n}{9}$ , then the value of  $n$  is:

OPTIONS: (A) 5 (B) 4 (C) 9 (D) 10

ANSWER: (C) 9

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YEAR: 2024

CHAPTER: 12 - STATISTICS

ORIGINAL\_SECTION: D

ORIGINAL\_MARKS: 5

QUESTION: The following table shows the ages of the patients admitted in a hospital during a year. Find the mode and mean of the data given above.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

---

YEAR: 2024

CHAPTER: 12 - STATISTICS

ORIGINAL\_SECTION: D

ORIGINAL\_MARKS: 5

QUESTION: The following distribution gives the monthly consumption of electricity of 68 consumers of a locality. Find the mean and mode of the data.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

---

YEAR: 2024

CHAPTER: 12 - STATISTICS

ORIGINAL\_SECTION: D

ORIGINAL\_MARKS: 5

QUESTION: The following distribution shows the daily pocket allowance of children of a locality. The mean daily pocket allowance is ₹ 36.10. Find the missing frequency, f.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

---

YEAR: 2024

CHAPTER: 12 - STATISTICS

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The mean of five numbers is 15. If we include one more number, the mean of six numbers becomes 17. The included number is:

OPTIONS: (A) 27 (B) 37 (C) 17 (D) 25

ANSWER: (A) 27

---

YEAR: 2024

CHAPTER: 12 - STATISTICS

ORIGINAL\_SECTION: D

ORIGINAL\_MARKS: 5

QUESTION: The following table shows the ages of the patients admitted in a hospital during a year. Find the mode and mean of the data given above.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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\*\*CHAPTER: 13 - PROBABILITY\*\*

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YEAR: 2024

CHAPTER: 13 - PROBABILITY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The number of red balls in a bag is 10 more than the number of black balls. If the probability of drawing a red ball at random from this bag is  $\frac{3}{5}$ , then the total number of balls in the bag is:

OPTIONS: (A) 50 (B) 60 (C) 80 (D) 40

ANSWER: (A) 50

---

YEAR: 2024

CHAPTER: 13 - PROBABILITY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: A pair of dice is thrown. The probability that sum of numbers appearing on top faces is at most 10 is:

OPTIONS: (A)  $\frac{1}{11}$  (B)  $\frac{10}{11}$  (C)  $\frac{5}{6}$  (D)  $\frac{11}{12}$

ANSWER: (D)  $\frac{11}{12}$

---

YEAR: 2024

CHAPTER: 13 - PROBABILITY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: A card is selected at random from a deck of 52 playing cards. The probability of it being a red face card is:

OPTIONS: (A)  $\frac{3}{13}$  (B)  $\frac{2}{13}$  (C)  $\frac{1}{2}$  (D)  $\frac{3}{26}$

ANSWER: (D)  $\frac{3}{26}$

---

YEAR: 2024

CHAPTER: 13 - PROBABILITY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The probability of getting a bad egg in a lot of 400 eggs is 0.045. The number of good eggs in the lot is:

OPTIONS: (A) 18 (B) 180 (C) 382 (D) 220

ANSWER: (C) 382

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YEAR: 2024

CHAPTER: 13 - PROBABILITY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: A bag contains 3 red balls, 5 white balls and 7 black balls. The probability that a ball drawn from the bag at random will be neither red nor black is:

OPTIONS: (A)  $\frac{1}{3}$  (B)  $\frac{1}{5}$  (C)  $\frac{7}{15}$  (D)  $\frac{8}{15}$

ANSWER: (A)  $\frac{1}{3}$

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YEAR: 2024

CHAPTER: 13 - PROBABILITY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The probability of getting a composite number greater than 3 on throwing a die is:

OPTIONS: (A)  $\frac{1}{6}$  (B)  $\frac{1}{3}$  (C)  $\frac{1}{2}$  (D)  $\frac{2}{3}$

ANSWER: (B)  $\frac{1}{3}$

---

YEAR: 2024

CHAPTER: 13 - PROBABILITY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: Two dice are thrown together. The probability that they show different numbers is:

OPTIONS: (A)  $\frac{1}{6}$  (B)  $\frac{5}{6}$  (C)  $\frac{1}{3}$  (D)  $\frac{2}{3}$

ANSWER: (B)  $\frac{5}{6}$

---

YEAR: 2024

CHAPTER: 13 - PROBABILITY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: If a digit is chosen at random from the digits 1, 2, 3, 4, 5, 6, 7, 8, 9; then the probability that this digit is an odd prime number is:

OPTIONS: (A)  $\frac{1}{3}$  (B)  $\frac{2}{3}$  (C)  $\frac{4}{9}$  (D)  $\frac{5}{9}$

ANSWER: (A)  $\frac{1}{3}$

---

YEAR: 2024

CHAPTER: 13 - PROBABILITY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: One ticket is drawn at random from a bag containing tickets numbered 1 to 40. The probability that the selected ticket has a number which is a multiple of 7 is:

OPTIONS: (A)  $\frac{1}{7}$  (B)  $\frac{1}{8}$  (C)  $\frac{1}{5}$  (D)  $\frac{7}{40}$

ANSWER: (C)  $\frac{1}{5}$

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YEAR: 2024

CHAPTER: 13 - PROBABILITY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: From the letters of the word "MOBILE", a letter is selected at random. The probability that the selected letter is a vowel, is:

OPTIONS: (A)  $\frac{3}{7}$  (B)  $\frac{1}{6}$  (C)  $\frac{1}{2}$  (D)  $\frac{1}{3}$

ANSWER: (D)  $\frac{1}{3}$

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YEAR: 2024

CHAPTER: 13 - PROBABILITY

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: A carton consists of 60 shirts of which 48 are good, 8 have major defects and 4 have minor defects. Nigam, a trader, will accept the shirts which are good but Anmol, another trader, will only reject the shirts which have major defects. One shirt is drawn at random from the carton. Find the probability that it is acceptable to Anmol.

OPTIONS: (A)  $\frac{4}{5}$  (B)  $\frac{1}{15}$  (C)  $\frac{2}{15}$  (D)  $\frac{13}{15}$

ANSWER: (A)  $\frac{4}{5}$

---

YEAR: 2024

CHAPTER: 13 - PROBABILITY

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: Three unbiased coins are tossed simultaneously. Find the probability of getting: (i) at least one head. (ii) exactly one tail. (iii) two heads and one tail.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

---

YEAR: 2024

CHAPTER: 13 - PROBABILITY

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: (a) Three coins are tossed simultaneously. What is the probability of getting (i) at least one head? (ii) exactly two tails? (iii) at most one tail?

OR

(b) A box contains 90 discs which are numbered 1 to 90. If one disc is drawn at random from the box, find the probability that it bears (i) a 2-digit number less than 40. (ii) a number divisible by 5 and greater than 50. (iii) a perfect square number.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 13 - PROBABILITY

ORIGINAL\_SECTION: C

ORIGINAL\_MARKS: 3

QUESTION: A jar contains 54 marbles, each of which is blue, green or white. The probability of selecting a blue marble at random from the jar is  $\frac{1}{3}$ , and the probability of selecting a green marble at random is  $\frac{4}{9}$ . How many white marbles does this jar contain?

OPTIONS: (A) 12 (B) 18 (C) 24 (D) 6

ANSWER: (A) 12

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YEAR: 2024

CHAPTER: 13 - PROBABILITY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The probability that a number selected randomly from the numbers 1, 2, 3, ..., 15 is a multiple of 4 is:

OPTIONS: (A)  $\frac{4}{15}$  (B)  $\frac{6}{15}$  (C)  $\frac{3}{15}$  (D)  $\frac{5}{15}$

ANSWER: (C)  $\frac{3}{15}$

---

YEAR: 2024

CHAPTER: 13 - PROBABILITY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: Two dice are tossed simultaneously. The probability of getting odd numbers on both the dice is:

OPTIONS: (A)  $\frac{6}{36}$  (B)  $\frac{3}{36}$  (C)  $\frac{12}{36}$  (D)  $\frac{9}{36}$

ANSWER: (D)  $\frac{9}{36}$

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YEAR: 2024

CHAPTER: 13 - PROBABILITY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: The probability of throwing a number greater than 2 with a fair die is:

OPTIONS: (A)  $\frac{2}{3}$  (B)  $\frac{1}{3}$  (C)  $\frac{1}{2}$  (D)  $\frac{5}{6}$

ANSWER: (A)  $\frac{2}{3}$

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YEAR: 2024

CHAPTER: 13 - PROBABILITY

ORIGINAL\_SECTION: B

ORIGINAL\_MARKS: 2

QUESTION: The king, queen and ace of clubs and diamonds are removed from a deck of 52 playing cards and the remaining cards are shuffled. A card is randomly drawn from the remaining cards. Find the probability of getting (i) a card of clubs. (ii) a red coloured card.

OPTIONS: Not an MCQ.

ANSWER: Not applicable.

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YEAR: 2024

CHAPTER: 13 - PROBABILITY

ORIGINAL\_SECTION: A

ORIGINAL\_MARKS: 1

QUESTION: For an event E, if  $P(E) + P(\text{not } E) = q$ , then the value of  $q^2 - 4$  is:

OPTIONS: (A) -3 (B) 3 (C) 5 (D) -5

ANSWER: (A) -3